

connected via Bluetooth to Alert Shirt app on their mobile phone. When viewers watch the game on TV, the jersey, helped by the app, makes them experience the same physical sensations impacting the selected players. It is basically haptic feedback on a larger scale.

Several years ago Philips demonstrated its Emotions Jacket, an experimental product that creates an immersive cinema experience. It allowed users to experience the intense emotions felt by on-screen characters. The jacket could simulate the sense of touch to create certain moods. This prototype can, if you think about it, be applied to other purposes, too. For example, you can sense when a baby is restless and then simulate the right emotions to help it settle down. You can also use a sense of touch to comfort and de-stress patients.

CuteCircuit has devised some such beautiful garments. One of the founders' first prototypes was Hug Shirt, which lets you send hugs over a distance. The sender's shirt has sensors that measure the strength, duration, location of touch, skin's warmth and heartbeat rate of the sender. The receiver's shirt has actuators that recreate the same hug, although they are a good distance apart. Of course, the shirt pairs with a mobile phone.

Recently, CuteCircuit developed a variant of this shirt, called Sound Shirt, which helps hearing impaired people experience music. Sound Shirt is packed with tiny actuators. It connects to a computer system, which picks up audio from microphones placed at various points on the stage, and makes the actuators cause a vibration according to the intensity of music being played in real time.

They used a bit of intuition to map the music to physical sensing. For example, deeper, heavier bass notes activate the actuators in lower parts of torso, and lighter notes like those produced by a violin activate the actuators further up on the body, around the neck area. As the wear-

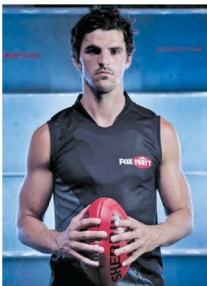


Fig. 5: Foxtel's Alert Shirt lets fans experience what their favourite football players feel on the field (Image courtesy: Wearable Experiments)

ers watch the orchestra and feel the pressure on certain areas, they gradually understand the correlation between actuation and notes, and start experiencing the music.

A bit of work remains to be done

All said, smart clothing is not without its hitches. Some problems still need to be ironed out before the mannequin in your neighbourhood store sports smart attire. Battery technology is still a real challenge. Truly stretchable and flexible batteries tend to be a bit expensive. And even then, their limited battery life requires the garments to be charged through universal serial bus or a power socket once in a while. This can get a little cumbersome for, say, a hiker who wants to wear a smartjacket and go on a long expedition; he or she might have trouble charging it.

So the industry is seriously exploring energy harvesting, which is energy from the Sun or from the wearer's movements, to power devices. Earlier attempts to put solar panels into fabrics have been a bit difficult—a bit stiff and not so good looking. Naturally, there is also a lot

more focus on miniaturisation and aesthetics. Several such prototypes are emerging.

Another real problem is how to make the smart garments more rugged. Nobody wants to buy a smartshirt and learn that it will last only 100 washes or so. In sultry and polluted environments such as those common in India, you have to wash the clothes frequently, which means these last much less. Plus, most of the current smart clothes cannot be dried in dryers because high temperatures as well as mechanical tumbling tend to spoil the electronics.

Then, there is the question of standards. Richard Poate, consumer products manager at TUV SUD Product Service, a global testing, certification, inspection and training provider, remarked in a media report that there are several problems with testing smart fabrics. As an example, he pointed out that there are numerous electrical standards for water and solid ingress. However, when it comes to smart textiles, available tests are not necessarily suitable. If we take the example of water ingress, available tests are for enclosures; these are not relevant to what smart textiles will experience.

He concluded by adding, "While technology moves quickly, standards tend to move painfully slowly. While there are general product directives and specific absorption rate tests, most of this technology relies on wireless technology, such as Bluetooth, to communicate. I would suggest developers to take their smart fabrics and treat these as these would be treated in the real world. Create your own tests."

Indeed, this situation, too, will change soon. Just like standards emerged for every other field of technology, these will emerge strong in the smart fabric space, too. It is a vicious cycle in the field of emerging tech. We wait for standards. Standards wait for adoption. And by some unsaid law of technological markets, everything ultimately works out fine! **EFY**